

Reminder: Empirical and simulation exercises must be submitted with “stand-alone” code and data (if any) in an appropriate format. The homework is due Sunday (data, code, and word-processed report by email to **the instructor before 10 PM Sunday**). Also, submit your work-processed report via **Gradescope**.

1. Let Ω be $\text{var}(e_t)$, where e_t is a two-dimensional reduced-form residual. Suppose that Ω is positive definite. Assume $e_t = D(0)v_t$, where v_t is a vector of structural shocks and $D(0)$ is a matrix. We further assume that $\text{var}(v_t) = I$.
 - a) Let C be a lower-triangular matrix with positive elements on its main diagonal such that $CC' = \Omega$. It is known that such C uniquely exists. Prove that C is invertible. HINT: $\text{rank}(AA') = \text{rank}(A)$ for any matrix A .
 - b) Denote $C^{-1}D(0)$ by R . Show that $RR' = I$. That is, R is a two-dimensional orthogonal matrix.
 - c) It is known that a two-dimensional orthogonal matrix can be represented as either $RO(\theta) = \begin{bmatrix} \cos(\theta) & -\sin(\theta) \\ \sin(\theta) & \cos(\theta) \end{bmatrix}$ or $RE(\theta) = \begin{bmatrix} \cos(\theta) & \sin(\theta) \\ \sin(\theta) & -\cos(\theta) \end{bmatrix}$. The former is a rotation matrix, while the latter is a reflection matrix. Show that $RE(\theta) = RO(\theta) \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$. Explain why we can assume that R is $RO(\theta)$ for some θ without loss of generality. Remember that our identification is up to signs.
2. Blanchard and Quah (AER, 1989) identify technology shocks using long-run restrictions. They estimate a bivariate VAR

$$\begin{bmatrix} \Delta y_t \\ z_t \end{bmatrix} = c + \Phi_1 \begin{bmatrix} \Delta y_{t-1} \\ z_{t-1} \end{bmatrix} + \dots + \Phi_p \begin{bmatrix} \Delta y_{t-p} \\ z_{t-p} \end{bmatrix} + e_t = \mu + \begin{bmatrix} D_{11}(L) & D_{12}(L) \\ D_{21}(L) & D_{22}(L) \end{bmatrix} \begin{bmatrix} v_t^d \\ v_t^s \end{bmatrix},$$
 where y is log of real GDP and z is the unemployment rate. Download these data but, in contrast to Blanchard and Quah (BQ), do not detrend any time series.
 - a) Using BIC, choose the lag length in the VAR and estimate it.
 - b) Using BQ's long-run restrictions, estimate matrix A as defined below

$$e_t = \begin{bmatrix} D_{11}(0) & D_{12}(0) \\ D_{21}(0) & D_{22}(0) \end{bmatrix} \begin{bmatrix} v_t^d \\ v_t^s \end{bmatrix} = Av_t,$$

where v^d is the demand side shock and v^s is the supply (technology) shock. Also, use the normalization that $\text{var}(v^s) = \text{var}(v^d) = 1, \text{cov}(v^s, v^d) = 0$. Interpret your estimates.

- c) Using point estimates, construct the impulse responses to supply and demand shocks at horizons $h=0, \dots, 50$. Discuss your results.
- d) Using point estimates, compute the variance decomposition at horizons $h=0, \dots, 50$. Discuss your results.
- e) Using bootstrap, compute 95% confidence bands for the impulse response.
- f) Now triple the number of lags in the VAR and repeat b)-e). Compare the differences.